SkillNavora

PROGRAMME GUIDE

AI Forward Deployed Engineer, Enterprise AI Architect & AI Automation Business Handbook

The role, four career tracks, the seven-stage roadmap, the volume plan, the tool directory and career links.

From Non-Coder to Enterprise AI Builder: Learn, Build, Deploy, Get Hired and Launch Your AI Business.

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Financial figures are hypothetical examples, not advice.

Programme guide

What a Forward Deployed Engineer is

> A Forward Deployed Engineer (FDE) works directly with customers to understand their problems, then builds, deploys and supports software and AI solutions on the customer's own systems and data, and owns the outcome. Palantir popularised the role; AI companies and consultancies such as OpenAI, Anthropic and Accenture now hire for it. It combines software engineering, AI engineering, solutions architecture, product thinking and consulting.

Role definitions vary between companies, so read each job description carefully. Compensation depends on skills, experience, location, employer and proven production work; this programme makes no salary or placement promise.

Dimension	Product Manager	AI Engineer	Forward Deployed Engineer
Understanding business problems	High	Medium	High
Product strategy	High	Medium	Medium–High
Coding	Usually limited	High	High
AI development	Conceptual / technical	High	High
Customer interaction	High	Variable	Very high
Deployment	Coordinates	Builds and deploys	Owns customer deployment
Business outcomes	Product outcomes	Technical outcomes	Deployment and business outcomes
Architecture	Product level	Technical	Enterprise and application
Integration	Coordinates	Implements	Designs and implements
Troubleshooting	Product issues	Technical issues	Customer and production issues

Who this programme is for

Absolute beginners, non-coders, working professionals, graduates, developers, analysts, product managers, consultants and entrepreneurs. It starts from zero and ends at production-ready, customer-facing work, with two outcomes: employment as an FDE or related role, and launching an AI automation agency or B2B AI SaaS.

Learning roadmap: seven stages

STAGE 1

Programming foundations

Topics: Computer and software basics, Python, SQL, Git, APIs, JavaScript basics, Terminal, Debugging

Outcome: Understand programming; build simple applications

STAGE 2

Applied AI

Topics: OpenAI API, Claude API (Anthropic), Gemini API (Google), Prompt engineering, Context engineering, Embeddings, Tokenisation, Basic RAG

Outcome: Build practical LLM-powered applications

STAGE 3

Agent development

Topics: LangChain, LangGraph, CrewAI, Tool calling, MCP (Model Context Protocol), Agent memory, Multi-agent workflows, Human-in-the-loop

Outcome: Build tool-using agents and automated workflows

STAGE 4

Application development

Topics: FastAPI, React, Frontend and backend, Database integration, Authentication, Authorisation, SaaS fundamentals

Outcome: Build complete AI-powered applications

STAGE 5

Production deployment

Topics: Cloud: AWS, Cloud: Microsoft Azure, Cloud: Google Cloud Platform (GCP), Docker, Kubernetes, CI/CD, Monitoring and observability, Security in production, AI evaluation, Production troubleshooting

Outcome: Deploy secure, scalable applications

STAGE 6

Enterprise projects

Topics: Real estate: AI lead response and property matching platform, Education: AI learning assistant and student support platform, Financial services: document processing and application review automation

Outcome: Demonstrate end-to-end project ownership

STAGE 7

Professional and entrepreneurial readiness

Topics: FDE interviews, Architecture presentations, Customer discovery, Client demos, Agency packaging, SaaS MVP, Sales

Outcome: Prepare for employment and independent business

Each stage has its own detailed handbook (Stage 1 to Stage 7 PDFs). The 14 module handbooks explain every topic in the curriculum.

Volume plan

1. Volume 1: Introduction to FDE and Career Foundations
2. Volume 2: Programming and Software Engineering
3. Volume 3: Data Engineering, SQL and Databases
4. Volume 4: AI and Machine Learning
5. Volume 5: Generative AI and LLMs
6. Volume 6: RAG and Knowledge Systems
7. Volume 7: AI Agents and Automation

8. Volume 8: Full-Stack Development
9. Volume 9: Cloud, DevOps and Deployment
10. Volume 10: AI Product Management and Business Thinking
11. Volume 11: Enterprise Architecture and Integration
12. Volume 12: Security, Governance and Responsible AI
13. Volume 13: AI Automation Consulting and Agency Building
14. Volume 14: B2B AI SaaS Development
15. Volume 15: Four End-to-End Industry Projects
16. Volume 16: Career Preparation and Interview Handbook
17. Volume 17: Tools, Glossary and Learning Resources

Four career tracks

Choose one track to focus on; the shared foundation is the same.

TRACK A

AI Forward Deployed Engineer

> Customer problems to production: Python apps, LLM integrations, RAG, agents, APIs, cloud deployment, debugging and monitoring.

WHO IT IS FOR

Developers, engineers and technically curious non-coders who enjoy building and working directly with customers.

THE ROLE

An FDE embeds with a customer, learns their process and systems, builds and deploys AI solutions on their infrastructure and data, and owns the outcome until it is adopted and measured. The common description is 'one customer, many capabilities', compared with a product engineer's 'one capability, many customers'.

A TYPICAL WEEK

- Discovery calls and process observation with customer teams
- Writing Python/TypeScript integrations, RAG pipelines and agents
- Connecting CRMs, ERPs, databases and document stores through APIs
- Deploying with Docker and CI/CD to AWS, Azure or GCP
- Building evaluation sets and monitoring dashboards
- Debugging production issues and presenting results to stakeholders

SKILLS TO BUILD

- Python, SQL, Git, APIs, JavaScript basics
- LLM APIs, prompt and context engineering, RAG, embeddings
- Agents, tool calling, MCP, human-in-the-loop
- FastAPI, React, authentication and authorisation
- Docker, Kubernetes basics, CI/CD, one cloud in depth
- Evaluation, observability, security, customer communication

MODULES TO PRIORITISE

1, 2, 4, 5, 6, 7, 8, 11, 12 (deep); 3, 9, 10 (working knowledge)

PORTFOLIO PROJECTS

- RAG assistant with citations and an evaluation pipeline
- Tool-using agent with approval gates deployed to the cloud
- One full industry project from Stage 6

JOB TITLES TO SEARCH

- Forward Deployed Engineer
- Forward Deployed AI Engineer
- AI Deployment Engineer
- Solutions Engineer (AI)
- Applied AI Engineer

WHAT INTERVIEWS TEST

- Practical Python and SQL
- Build a small LLM feature in a few hours
- AI system design (RAG, agents, integration)
- Customer scenario role-play
- STAR stories about ownership and ambiguity

90-DAY PLAN

- Days 1–30: Stage 1 and Stage 2; ship a deployed RAG demo
- Days 31–60: Stages 3–4; ship an agent with tools, auth and a React UI
- Days 61–90: Stage 5 and one Stage 6 project; prepare interviews (Stage 7)

TRACK B

AI Solutions Architect

> Enterprise architecture, multi-provider AI design, integration layers, security, governance, cost estimates and architecture documents.

WHO IT IS FOR

Experienced engineers, architects, cloud and integration specialists moving into AI.

THE ROLE

An AI solutions architect designs how AI capabilities fit into an organisation's systems: model providers, gateways, retrieval, agents, integration, identity, security, governance, reliability and cost. They write architecture documents and decision records, lead reviews, and guide delivery teams.

A TYPICAL WEEK

- Requirement and constraint workshops with IT, security and business
- Drawing context, container and data-flow diagrams
- Choosing providers, hosting and integration patterns; writing ADRs
- Cost modelling and capacity planning
- Security and compliance reviews with CISOs
- Reviewing implementations and guiding teams

SKILLS TO BUILD

- Enterprise architecture and integration patterns
- Cloud architecture on AWS, Azure and GCP
- Multi-provider LLM design, AI gateways, model routing
- Enterprise RAG with permissions; agent orchestration
- Security, governance, responsible AI, compliance
- Cost estimation, SLOs, reliability, documentation

MODULES TO PRIORITISE

8, 10, 12 (deep); 4, 5, 6, 9, 11 (strong); 1, 2, 7 (working knowledge)

PORTFOLIO PROJECTS

- Reference architecture for an enterprise AI platform with an AI gateway
- Architecture document and ADRs for one Stage 6 project
- Cost and risk model comparing three providers and two hosting options

JOB TITLES TO SEARCH

- AI Solutions Architect
- Enterprise AI Architect
- Cloud AI Architect
- GenAI Architect
- Principal Solutions Architect

WHAT INTERVIEWS TEST

- Architecture design exercise with trade-offs
- Security and governance scenarios

- Cost estimation
- Presenting to executives
- Past architecture decisions and their outcomes

90-DAY PLAN

- Days 1–30: Modules 4–6 hands-on (build RAG and an agent)
- Days 31–60: Modules 8, 10, 12; write a reference architecture
- Days 61–90: Architecture presentations, ADRs, cost models; interview practice

TRACK C

AI Product Manager

> Opportunity discovery, feasibility, roadmaps, PRDs, MVPs, AI metrics, launches, adoption and ROI.

WHO IT IS FOR

Product managers, business analysts, consultants and domain experts moving into AI products.

THE ROLE

An AI product manager decides which AI products and features to build, for whom and why; defines requirements including quality targets and evaluation; works with engineering and design to ship; and drives adoption and measurable business impact responsibly.

A TYPICAL WEEK

- Customer and user discovery
- Assessing AI opportunities and feasibility with engineers
- Writing PRDs with AI quality, latency and cost requirements
- Defining evaluation sets and success metrics
- Planning launches, onboarding and change management
- Tracking adoption, feedback and ROI

SKILLS TO BUILD

- Product discovery, JTBD, process mapping
- AI fundamentals, LLM capabilities and limits, RAG and agents concepts
- PRDs, user stories, acceptance criteria, roadmaps
- AI evaluation and metrics, AI UX, responsible AI
- Analytics, adoption, business cases and ROI
- Stakeholder management and communication

MODULES TO PRIORITISE

9, 11 (deep); 3, 4, 5, 6, 12 (strong conceptual plus small builds); 1, 2, 7, 13, 14 (working knowledge)

PORTFOLIO PROJECTS

- PRD and evaluation plan for an AI feature, with a no-code prototype (Lovable)
- Business case and ROI model for one Stage 6 project
- Launch and adoption plan with metrics dashboard

JOB TITLES TO SEARCH

- AI Product Manager
- Product Manager, GenAI
- Technical Product Manager (AI)
- AI Product Owner
- Product Lead, Applied AI

WHAT INTERVIEWS TEST

- Product sense for AI features
- Metrics and evaluation design

- Prioritisation and trade-offs
- Working with engineering on feasibility
- Responsible AI scenarios

90-DAY PLAN

- Days 1–30: Modules 3–5 concepts plus a Lovable prototype
- Days 31–60: Module 9; PRD, evaluation plan and business case
- Days 61–90: Adoption and launch plan; portfolio and interviews

TRACK D

AI Automation Consultant

> Process audits, no-code and AI workflows, CRM and ERP integration, support and document automation, selling and supporting solutions.

WHO IT IS FOR

Freelancers, operations professionals, marketers, entrepreneurs and non-coders who want to run an AI automation practice.

THE ROLE

An AI automation consultant audits business processes, designs and builds automations with no-code tools, AI models and light code, integrates them with the client's apps, and sells and supports them through packaged services and retainers.

A TYPICAL WEEK

- Discovery calls and automation audits
- Mapping workflows and estimating ROI
- Building automations in n8n, Zapier or Make with AI steps
- Integrating CRM, email, WhatsApp, documents and calendars
- Testing, training clients and monitoring runs
- Prospecting, proposals and account management

SKILLS TO BUILD

- Process mapping and ROI estimation
- n8n, Zapier, Make, webhooks, APIs
- Prompting, structured outputs, basic RAG and agents
- CRM, email, WhatsApp and document automation
- Data privacy and safe automation practices
- Packaging, pricing, sales and client communication

MODULES TO PRIORITISE

6.7, 13 (deep); 2.4, 4, 5, 9, 11 (strong); 1, 12, 14 (working knowledge)

PORTFOLIO PROJECTS

- Lead response automation for a real-estate agency
- Support assistant from a knowledge base with escalation
- Document extraction workflow into a spreadsheet or CRM

JOB TITLES TO SEARCH

- AI Automation Consultant
- Automation Specialist
- AI Agency Founder
- Workflow Automation Engineer
- No-code AI Developer

WHAT INTERVIEWS TEST

- Portfolio walkthrough of live automations

- Live workflow build
- ROI and pricing questions
- Client scenario role-play

90-DAY PLAN

- Days 1–30: Prompting, APIs, n8n basics; build two automations
- Days 31–60: RAG support bot, CRM and WhatsApp integrations; first paid audit
- Days 61–90: Package offers, three case studies, outreach pipeline

Tool directory: what each tool is and what it is for

You do not need every tool. Learn one option per category first, then compare.

Category	Tool	What it is	Used for	Cost model	Link
Programming	**Python**	General-purpose language and the default for AI, data and automation.	Scripts, APIs (FastAPI), data processing, model SDKs, agents.	Free, open source	docs
Programming	**JavaScript**	The language of web browsers and Node.js servers.	Interactive front ends, small backends, browser automation.	Free	docs
Programming	**TypeScript**	JavaScript with static types, compiled to JavaScript.	Larger front ends (React, Next.js) and Node backends with fewer bugs.	Free, open source	docs
AI models	**OpenAI API**	API for OpenAI's GPT models (text, reasoning, vision, audio, embeddings).	Chat, extraction, agents, embeddings; also via Azure OpenAI.	Pay per token	docs
AI models	**Claude API (Anthropic)**	API for Anthropic's Claude models through the Messages API.	Long documents, coding, analysis, tool use and agents; also on Bedrock, Vertex AI and Foundry.	Pay per token	docs
AI models	**Gemini API (Google)**	API for Google's Gemini multimodal models via AI Studio or Vertex AI.	Multimodal input (images, audio, video), long context, Google Cloud integration.	Free tier for development; pay per token	docs
AI coding	**VS Code**	Free, extensible code editor from Microsoft.	Editing, debugging, Git, extensions for Python, Docker, Kubernetes.	Free	docs
AI coding	**Cursor**	AI-first code editor built on VS Code with chat and multi-file edits.	Writing and refactoring code with AI assistance across a repository.	Free tier; paid plans	docs
AI coding	**GitHub Copilot**	AI pair programmer inside editors and GitHub.	Inline suggestions, chat about code, pull request summaries.	Free tier; paid plans	docs
AI coding	**Claude Code**	Anthropic's agentic coding tool in the terminal, IDE, desktop and web.	Reading repositories, editing files, running commands and tests with permission.	Included with Claude plans or API billing	docs
Application builders	**Lovable**	AI app builder that generates React web apps from prompts and connects to Supabase.	Prototypes, MVPs and internal tools without writing everything by hand.	Free tier; paid plans	docs

AI frameworks	**LangChain**	Framework of components for LLM apps: models, prompts, tools, retrievers, loaders.	Provider-agnostic chains, RAG and integrations.	Free, open source	docs
AI frameworks	**LangGraph**	Library for stateful agent workflows as graphs with persistence and human-in-the-loop.	Controllable production agents and multi-step workflows.	Free, open source	docs
AI frameworks	**CrewAI**	Framework for role-based multi-agent crews and flows.	Quick multi-agent prototypes such as research and report teams.	Free, open source; paid platform	docs
AI frameworks	**LlamaIndex**	Data framework for ingesting, indexing and querying documents with LLMs.	RAG pipelines, document parsing and data agents.	Free, open source; paid cloud	docs
AI agent platforms	**OpenAI Agents SDK**	Lightweight library for agents, tools, hand-offs, guardrails and tracing.	Building agents on OpenAI models in Python or TypeScript.	Free, open source (model usage billed)	docs
AI agent platforms	**Google Agent Development Kit (ADK)**	Google's framework for building, evaluating and deploying agents.	Agents on Gemini and Vertex AI, multi-agent systems.	Free, open source (model usage billed)	docs
Databases	**PostgreSQL**	Open-source relational database with strong SQL support and extensions.	Application data, analytics queries, vectors with pgvector.	Free, open source	docs
Databases	**Supabase**	Hosted PostgreSQL with authentication, storage, edge functions and vector search.	Fast backends for MVPs and SaaS; pairs with Lovable.	Free tier; paid plans	docs
Databases	**MySQL**	Widely used open-source relational database.	Web applications and many existing customer systems.	Free community edition	docs
Databases	**MongoDB**	Document database storing JSON-like records.	Flexible schemas, content and event data; Atlas adds vector search.	Free community edition; Atlas free tier	docs
Vector databases	**Pinecone**	Fully managed, serverless vector database.	Large-scale semantic search and RAG without running infrastructure.	Free starter tier; usage-based	docs
Vector databases	**Qdrant**	Open-source vector database with rich filtering.	Self-hosted or cloud semantic search and RAG.	Free open source; cloud free tier	docs

Vector databases	**pgvector**	PostgreSQL extension adding vector columns and similarity search.	Keeping vectors next to application data in PostgreSQL.	Free, open source	docs
APIs	**Postman**	Platform for building, testing and documenting APIs.	Testing endpoints, sharing collections, mock servers.	Free tier; paid plans	docs
APIs	**FastAPI**	Modern Python web framework for typed, async APIs with automatic docs.	Backends for AI features and integrations.	Free, open source	docs
Version control	**Git**	Distributed version control system.	Tracking changes, branching, merging and collaboration.	Free, open source	docs
Version control	**GitHub**	Hosting for Git repositories with pull requests, issues, Actions and security tools.	Collaboration, code review, CI/CD and portfolios.	Free tier; paid plans	docs
Cloud	**AWS**	Amazon's cloud platform: compute, storage, databases, networking, Bedrock for AI.	Hosting and scaling applications in the customer's AWS account.	Free tier; pay as you go	docs
Cloud	**Microsoft Azure**	Microsoft's cloud with Entra ID integration and Azure AI Foundry.	Enterprise deployments in Microsoft-centric organisations.	Free account credits; pay as you go	docs
Cloud	**Google Cloud (GCP)**	Google's cloud with Cloud Run, BigQuery, GKE and Vertex AI.	Serverless containers, data analytics and Gemini/Claude on Vertex.	Free tier and credits; pay as you go	docs
Deployment	**Docker**	Packages applications into portable container images.	Consistent local development and cloud deployment.	Free for individuals and small businesses; paid tiers	docs
Deployment	**Kubernetes**	Open-source container orchestration system.	Running, scaling and updating containers across clusters.	Free, open source (managed services billed)	docs
Infrastructure	**Terraform**	Infrastructure as Code tool using HCL to define cloud resources.	Repeatable, reviewable cloud environments across providers.	Free CLI; paid cloud platform	docs
Infrastructure	**Helm**	Package manager for Kubernetes applications (charts).	Templated, versioned Kubernetes deployments per environment.	Free, open source	docs
CI/CD	**GitHub Actions**	Workflow automation built into GitHub.	Testing, building, evaluating and deploying on every change.	Free minutes; paid beyond	docs

Automation	**n8n**	Workflow automation with visual nodes, code nodes and AI agent nodes; self-hostable.	Complex automations and AI workflows with data control.	Free self-hosted community edition; paid cloud	docs
Automation	**Zapier**	No-code automation connecting thousands of apps.	Quick automations for non-technical teams.	Free tier; paid plans	docs
Automation	**Make**	Visual automation platform with detailed data mapping.	Multi-step scenarios with branching and transformations.	Free tier; paid plans	docs
Product management	**Jira**	Work tracking for agile teams.	Backlogs, sprints, bugs and roadmaps.	Free tier; paid plans	docs
Product management	**Confluence**	Team documentation and knowledge base.	PRDs, architecture documents, runbooks and meeting notes.	Free tier; paid plans	docs
Design	**Figma**	Collaborative interface design and prototyping tool.	Screens, design systems and clickable prototypes.	Free tier; paid plans	docs
Design	**FigJam**	Figma's online whiteboard.	Workshops, process maps, user journeys and brainstorming.	Free tier; paid plans	docs
Design	**Miro**	Visual collaboration whiteboard.	Discovery workshops, mapping and planning with clients.	Free tier; paid plans	docs
Analytics	**SQL**	Query language for relational databases.	Analysis, reporting and data checks.	Free (part of databases)	docs
Analytics	**Power BI**	Microsoft's business intelligence and dashboard tool.	Executive dashboards and ROI reporting.	Free desktop; paid sharing	docs
Analytics	**Mixpanel**	Product analytics for events, funnels and retention.	Measuring activation, feature adoption and retention.	Free tier; paid plans	docs
Monitoring	**OpenTelemetry**	Open standard and tools for metrics, logs and traces.	Instrumenting apps and AI calls; exporting to any monitoring backend.	Free, open source	docs

Recommended stacks

BEGINNER STACK

Python, VS Code, Git/GitHub, one LLM API (Claude, OpenAI or Gemini), SQLite or Supabase, Lovable for UI, n8n or Zapier for automation, Postman.

INTERMEDIATE STACK

Python + FastAPI, React + TypeScript, PostgreSQL + pgvector, LangGraph or plain SDK tool loops, MCP, Docker, GitHub Actions, one cloud (Cloud Run, App Runner or Container Apps), basic logging and

evaluation scripts.

ADVANCED STACK

Multi-provider LLM gateway, enterprise RAG with permission filtering and reranking, LangGraph agents with checkpoints and approvals, Kubernetes with Helm, Terraform, OIDC-based CI/CD, OpenTelemetry tracing with LLM observability, automated evaluation gates, secrets management and red-team testing.

Career preparation: job links

Use current postings to compare required skills with this curriculum. Postings open and close often; confirm each role is still open.

Organisation	Role	Link
OpenAI	Forward Deployed Engineer	https://openai.com/careers/search/
OpenAI	Forward Deployed Engineering roles (search)	https://openai.com/careers/search/?q=forward
Palantir	Forward Deployed AI Engineer	https://jobs.lever.co/palantir/ff1029bd-bb6d-4d78-a03e-5f9744d0b798
Accenture India	Forward Deployed Engineer	https://www.accenture.com/in-en/careers/jobdetails?id=R00344052_en
Accenture India	Forward Deployed AI Engineer	https://www.accenture.com/in-en/careers/jobdetails?id=ATCI-5613009-S2028500_en

How to use these handbooks

1. Read the Programme guide and pick a track.
2. Work through the stage handbooks in order; do every exercise.
3. Use the module handbooks as the reference for any topic.
4. Build the three Stage 6 projects at the level that suits you (beginner, intermediate, advanced).
5. Use Stage 7 to prepare interviews, demos, offers and sales.